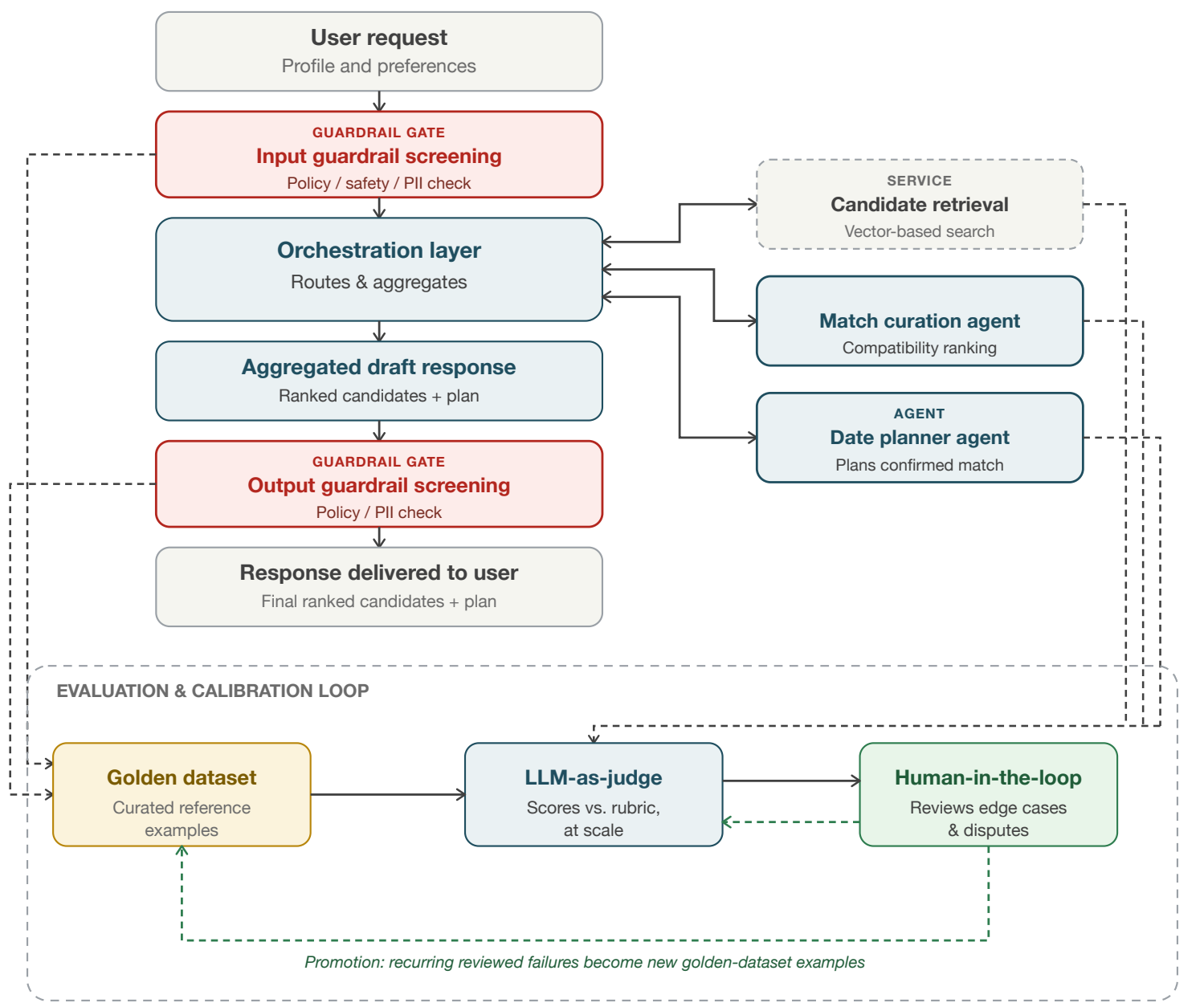


Agentic AI Evaluation & Architecture Overview

Match Curation Agent & Date Planner Agent — Stealth AI B2C Marketplace
Nainy Mehra | Founding Product Manager

Designed this AI evaluation framework to assess production readiness for the Agentic Dating app (stealth project). Also submitted as part of the team's YC application.

1. System Architecture: Orchestration, Agents, Guardrail Gates & Evaluation/Calibration Loop



2. Evaluation Layers — Metrics by Component

2a. Guardrail gates (outer loop)

Gate	Offline metrics	Online metrics
------	-----------------	----------------

Input guardrail screening Before any agent	Violation catch rate, false-positive rate, added latency	Escalation rate, override rate
Output guardrail screening After agents, before user	Violation catch rate, false-negative rate, added latency	Escalation rate, override rate, incident rate

Checked once per request/response at the boundary — not duplicated inside every agent.

2b. Per-agent task & quality evals (inner loop)

Component	Task execution (offline)	Output quality (offline)	Business outcome (online)
Candidate retrieval Vector-based search	Tool-call success rate	Recall@k, Precision@k	Latency (P95), conversion contribution
Match curation agent Compatibility ranking	Task completion rate	Precision, Recall, NDCG, Satisfaction	Acceptance rate, conversion lift, retention
Date planner agent Plans confirmed match	Tool-call success, task completion	Feasibility rate, budget fit	Booking rate, post-date satisfaction

Scored offline against the golden dataset; business outcome is the only online metric.

3. Golden Dataset Preparation

Step	What happens
Source	~60% real anonymized interactions, ~40% synthetic edge cases (safety scenarios, conflicting preferences)
Stratify	Every example tagged by difficulty (clear-fit / ambiguous / adversarial) and safety tier (standard / escalation-risk)
Weight	Hard, rare, high-risk cases are overweighted on purpose — the set optimizes for catching failures, not mirroring real traffic
Label	Each example gets a human-rater expected output + rubric score, becoming ground truth
Version	Frozen per release (v1, v2...) with a held-out 10% slice never used for tuning — only for final validation

Deliberately skewed toward hard and rare cases — it's built to catch failures, not to represent everyday traffic.

4. Feedback Loop: How the System Self-Improves

Step	What happens
Capture	Every user override, low-confidence judge score, or rejected output is logged as a failure signal
Tag	Each failure gets one primary category: guardrail / matching / planning / trust-UX / escalation
Cluster	Weekly, failures are grouped by category and similarity — highest-frequency patterns surface first
Promote	A cluster of 3+ similar failures becomes a new golden-dataset candidate, human-labeled and added
Re-validate	The next release is tested against the expanded set — graded against real failures it produced, not just original guesses

Production failures become permanent test cases — the system can't fail the same way twice without it surfacing in every

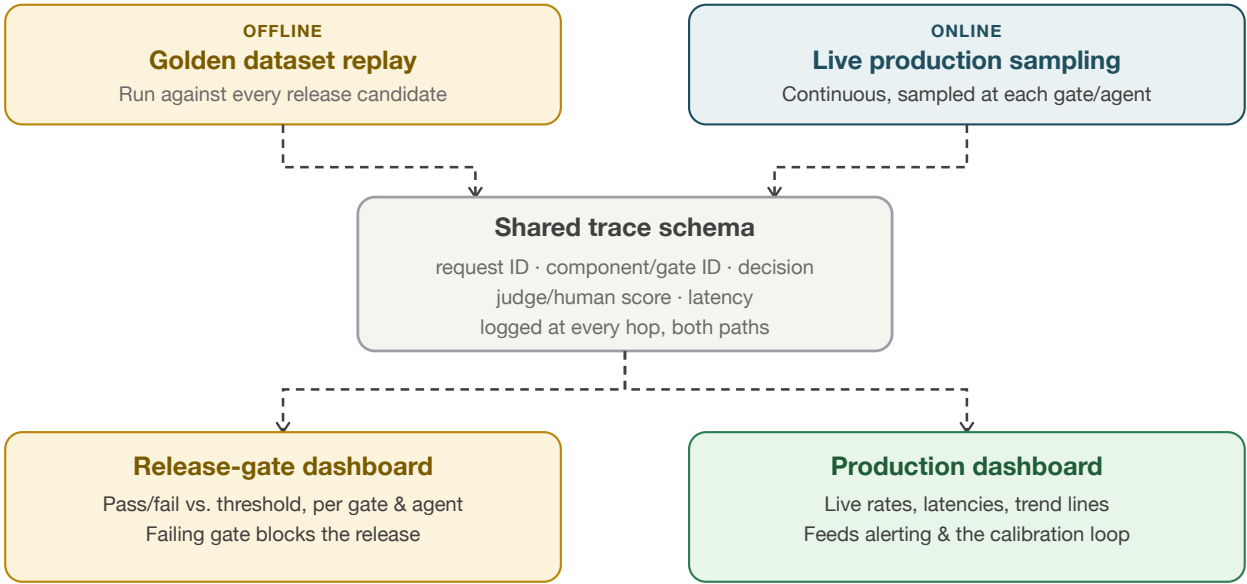
future release check.

4a. Failure taxonomy (five categories)

Category	Definition	Example trigger	Severity	Routing
Guardrail failure	Policy, safety, or privacy boundary crossed	Unflagged red-flag profile pattern surfaced	Critical	Immediate block + human review
Matching failure	Curated match doesn't reflect preferences	Match contradicts an explicit dealbreaker	High	Golden-set candidate + prompt review
Planning failure	Date plan infeasible or budget-inappropriate	Suggested venue closed at proposed time	Medium	Golden-set candidate
Trust / UX failure	Technically correct but erodes confidence	Plan lacks any inspectable rationale	Medium	UX backlog + golden-set candidate
Escalation failure	Agent proceeds where it should defer	Safety-sensitive scenario not routed for review	Critical	Immediate block + human review

Precedence rule when multiple categories apply: Guardrail/Escalation > Matching/Planning > Trust/UX — safety-relevant categories always take precedence.

5. Trace Flow: From Live Request to Dashboard



Achieved figures (68% match-relevance, 32% conversion lift) reflect early pilot results already shared publicly on this project. All thresholds shown elsewhere in the full framework are release-gate targets, not additional measured claims.